

# ArchiLudo

A Ludo (“Mens Erger Je Niet”) game for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. By Xander Mol (idreamtin8bits.com).

## Contents

- Playing ArchiLudo
- Installation
- Building from source
- Changelog
- Development history
- Documentation
- Credits and licence

## Playing ArchiLudo

Four players, four pawns each, a shared ring and a short home run per player – the classic “Mens Erger Je Niet”/Ludo cross board.

**Starting a game:** on first launch, dismissing the startup splash screen (**OK**, or click anywhere on it) opens the **New Game** dialogue directly. Later on, click the iconbar icon to open it again. Set each of the 4 players’ name and whether they’re Human or AI-controlled; an AI-controlled player also gets a **Low/Medium/ High** difficulty picker (click to cycle) – see AI.md for what each tier actually does. Optionally click **Rules...** to pick a rule variant or adjust individual house rules (see RULES.md for the full manual), then **Start**.

**Playing a turn:** click **Throw** to roll. The status line tells you what to do next – click one of your own pawns on the board if more than one can legally move (if only one can, it moves automatically), or click **Throw** again on a six. Landing exactly on another pawn sends it home; reaching the exact end of your home column finishes that pawn. For an AI player’s turn, click **Continue** to step through its moves. Every time a player finishes, a win screen appears – **Continue** (offered for 1st, 2nd, and 3rd place) lets the remaining players keep going to decide the rest of the placing; the 4th and final player’s screen has no Continue, since nobody’s left to play. Every win screen also offers **New Game** and **Quit Game**.

**Saving and loading:** **Save Game/Load Game** from the iconbar menu open 5 fixed, renamable slots at any time during play.

**Music and sound effects:** open the **Music** submenu (iconbar or window menu) to turn music **On/off**, turn **SFX** on/off independently, or pick a **Track**. See QTM.md for the full music/SFX manual (what each track and effect is, and Track 3’s one limitation).

**Rules and variants:** ArchiLudo ships 3 rule presets (Mens Erger Je Niet, Ludo, Pachisi-style) and 8 individually-adjustable house-rule toggles. See RULES.md

for the complete rules manual.

## Installation

**Requires a real or emulated Acorn Archimedes running RISC OS 3.10 or later** (ARM2 or ARM3). Download a release (**ArchiLudo-vX.Y.Z-\*.zip** or **.adf**) and:

- **Zip**: extract with SparkFS (or any RISC-OS-aware zip tool) – filetypes are preserved. Double-click the extracted **!ArchiLudo** application directory to run.
- **Disc image** (.adf, ADFS “D” format, 800KB): write it to a real 3.5” floppy, or mount it directly in an emulator that supports raw ADFS images. It contains just the release zip, correctly filetype – extract it the same way as above once it’s accessible from RISC OS.
- **Real classic-Econet hardware** (a PiEconetBridge or similar old-style fileserver): rename the downloaded file to 10 characters or fewer with no dot before transferring it (e.g. **ArchiZip** for the zip, **ArchiADF** for the disc image) – such filesevers silently truncate longer names in a way that makes the file unreadable, not just renamed. Arculator’s hostfs and a plain download/extract on Windows/Mac/Linux have no such limit.

No installer is needed beyond copying the **!ArchiLudo** directory wherever you want it – RISC OS applications are self-contained directories, not registry entries or system-wide installs.

## Building from source

### Prerequisites

| Tool      | Purpose                                                  | Install                                                                                                                                          |
|-----------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ArchieSDK | ARM2-targeting cross-compiler (GCC 8.5.0), bundled OSLib | <pre>git clone https://gitlab.com/_targz/archiesdk.git cd ~/riscos-dev/archiesdk ./build.sh</pre>                                                |
| Arculator | Acorn Archimedes emulator, for testing                   | <pre>install anywhere; point ARCULATOR_HOSTFS in .env at its hostfs folder (e.g. D:\Retro\Acorn\Arculator_V2.2_Windows\hostfs) on Windows)</pre> |
| pandoc    | optional: README.md -> README.pdf                        | <pre>sudo apt install pandoc texlive-xetex</pre>                                                                                                 |

| Tool              | Purpose                                                                    | Install                    |
|-------------------|----------------------------------------------------------------------------|----------------------------|
| Python 3 + Pillow | tools/riscos_sprite.py, pip install Pillow<br>the PNG->Sprite<br>converter |                            |
| sshpas            | optional: make<br>deploy-pibridge<br>(real-hardware deploy)                | sudo apt install<br>sshpas |

### .env setup

Copy .env.example to .env and set:

```
ARCHIESDK = /home/xahmol/riscos-dev/archiesdk
ARCULATOR_HOSTFS = /mnt/d/Retro/Acorn/Arculator_V2.2_Windows/hostfs
```

PIBRIDGE\_USER/PIBRIDGE\_HOST/PIBRIDGE\_PASS/PIBRIDGE\_PATH are also needed for make deploy-pibridge (a real PiEconetBridge target) – see .env.example for the full set. .env is gitignored – never commit it.

.env is read only by make (a plain -include .env in the Makefile), not by your shell – if you use VS Code, .vscode/c\_cpp\_properties.json points its IntelliSense compilerPath at \${env:ARCHIESDK}, which needs ARCHIESDK exported as a real environment variable separately (e.g. export ARCHIESDK=/path/to/archiesdk in your shell profile, or launch code . from a shell where you’ve already exported it).

### Make targets

| Target               | Effect                                                                                          |
|----------------------|-------------------------------------------------------------------------------------------------|
| make / make all      | build the build/!ArchiLudo application directory                                                |
| make test            | build and run the game-logic unit tests with the host compiler (no ArchieSDK/Arculator needed)  |
| make clean           | remove build/                                                                                   |
| make deploy          | copy build/!ArchiLudo to the Arculator hostfs folder                                            |
| make deploy-pibridge | deploy to a real PiEconetBridge over SSH (password auth)                                        |
| make zip             | versioned release archive (build/ArchiLudo-vX.Y.Z-<timestamp>.zip), RISC OS filetypes preserved |
| make disk            | ADFS “D” format (800KB) disc image containing the release zip, correctly filetype               |

| Target                                                 | Effect                                                                |
|--------------------------------------------------------|-----------------------------------------------------------------------|
| <code>make assets</code>                               | regenerate the pawn sprites and app icon from their Python generators |
| <code>make export-sprites / make import-sprites</code> | round-trip shipped sprites to hand-editable PNGs and back             |
| <code>make docs</code>                                 | regenerate <code>README.pdf</code> via pandoc                         |
| <code>make asm</code>                                  | emit generated ARM assembly for the current sources                   |

See the “Installation” section above for the 10-character filename limit on real classic-Econet hardware – `make zip/make disk` deliberately keep the full version+timestamp in their own output filenames so multiple local builds can be told apart.

Testing: boot `configs/ArchiLudo-ARM3-4MB.cfg` (matches real ARM3/4MB hardware) or `configs/ArchiLudo-ARM2-1MB.cfg` (stock ARM2/1MB compatibility check) in Arculator with the RISC OS 3.10 ROM, then double-click **!ArchiLudo** from HostFS via the Filer (see `BUILDCHAIN.md`’s “Application directory” section for its structure).

## Changelog

ArchiLudo hasn’t cut discrete numbered releases yet – this is a milestone-level history of what’s been built, newest first.

- **Documentation restructuring:** a new `TOOLS.md` manual now covers every standalone `tools//assets/*.py` script (usage, requirements, gotchas) in one place; `GRAPHICS_TOOLING.md`’s sprite-format/rendering-approach content moved into `ARCHITECTURE.md` instead of staying a separate file, and `BUILDCHAIN.md` now points to `TOOLS.md` rather than duplicating each script’s own mechanics.
- **Cleanup:** removed dead code – a vestigial, never-actually-used sample-format conversion path in `lib/qtm.c` and the 6 one-shot SFX files it was the only reason to ship separately in the app directory (embedded into the music tracks’ own MOD sample tables now, not shipped standalone); and a whole superseded pawn/dice art generator (`assets/generate_placeholder_art.py`) whose output nothing actually loaded any more, found while writing `TOOLS.md`. Debug file-logging (`debug_log()`) is now an opt-in build flag (`make DEBUG_LOG=1`) rather than always compiled in, so a default build carries none of its tracing strings or per-call file I/O.
- **Captured-pawn animation:** a displaced pawn (capture, own-pawn collision) now slides back to its home base instead of teleporting there instantly; the capture SFX was also replaced with a more expressive sound after two earlier tries read as too quiet or too flat in live testing – see

QTM.md.

- **Full placement flow:** a win screen (with fanfare) now appears for every player who finishes, not just the first – “WINS!” for 1st, “ended 2nd/3rd/4th” for the rest, each offering Continue (except the last), New Game, and Quit Game.
- **AI difficulty tiers:** Low/Medium/High per AI-controlled player, picked in the New Game dialogue – Low takes only the objectively decisive moves (win/finish/capture/progress), Medium is the original full heuristic, High adds a genuine one-ply lookahead. See AI.md.
- **Launch flow:** the New Game dialogue now opens automatically right after dismissing the startup splash screen.
- **Distribution pipeline:** a filetype-preserving release zip (`make zip`), a from-scratch ADFS disc-image builder (`make disk`), and a real-hardware deploy target over Econet (`make deploy-pibridge`) – all verified against real hardware, not just Arculator, including finding and fixing a 10-character filename limit on classic-Econet file servers and a line-ending mismatch in the bundled plain-text README.
- **QTM audio:** background music (3 selectable tracks) and 6 one-shot sound effects, embedded as MOD instrument samples and triggered via `QTM_PlaySample`, independently toggleable, with loudness-normalized SFX. Confirmed working on real hardware.
- **Save/load rework:** from a PRM-correct but non-functional (on Arculator’s HostFS) drag-and-drop design to 5 fixed, renamable save slots.
- **Multi-rule-set / house-rule variant system:** 3 curated presets (Mens Erger Je Niet, Ludo, Pachisi-style) plus 8 independently toggleable house rules, a Rule Options dialogue, and AI support for all of them – see RULES.md.
- **Win/continue flow:** a win no longer ends the game outright – players can continue for runner-up order or start a new game.
- **Sprite art pivot:** from `os_plot` primitives to real `Wimp_PlotIcon`-based pawn sprites and a proper application icon, plus a hand pixel-editing round-trip workflow for touching up art in an external editor.
- **AI opponents and player setup:** a “New Game” dialogue (names, Human/AI toggle per player) and a first AI opponent adapted from the original GeoLudo edition’s own algorithm.
- **Animation and redraw overhaul:** flicker-free small-region animation (dice roll, pawn slide, pulsing highlights) via careful `Wimp_UpdateWindow` usage.
- **Application directory packaging:** a real `!ArchiLudo` app directory with a custom icon, replacing an earlier flat `hostfs` layout.
- **Phase 1: playable core loop:** the WIMP game window, the real Mens Erger Je Niet board (ported from the original GEOS edition’s own coordinate tables), click-to-move, dice, capture/win detection.
- **Build environment:** the ArchieSDK toolchain, Arculator test profiles, and this project’s full documentation set established.

## Development history

ArchiLudo isn't this game's first outing – it's the latest in a line of ports going back over three decades, all by the same author. Full credits and technical detail for each generation are in `CREDITS.md`'s "Porting source" section; this is the narrative version.

- **1992:** the original, written in Commodore BASIC 7.0 for the Commodore 128 in 80-column mode – "Mens Erger Je Niet" (Dutch for "Don't Get Angry, Man", the house-rule variant of Ludo this whole family of ports implements), by Xander Mol under the "XAMA Software" name. The listing itself still carries its original 1992 copyright banner. Source and a working D64 disk image are preserved at `C128/BASIC Original` in the `xahmol/ludo` repository.
- **2020:** rewritten for the Oric Atmos, in BASIC.
- **2021:** rewritten a further three times – the Oric Atmos version again, now in C via CC65; a new C port for the TI-99/4a via TMS9900-GCC; and a new C port for the Commodore 128 via CC65.
- **2023: GeoLudo**, a GEOS edition sharing one executable across GEOS on the Commodore 64, Commodore 128, and Commodore Plus/4 – `xahmol/ludo/GEOS`. This is ArchiLudo's own direct porting source: its board geometry and ring/home-column layout, its pawn and die-face artwork, and its first AI opponent all trace back to GeoLudo specifically (see `docs/ARCHITECTURE.md`'s GeoLudo-to-ArchiLudo mapping table), even though ArchiLudo's own rules engine (`src/game_logic.c`) is a clean reimplementation rather than a line-for-line port (see `docs/GAME_LOGIC.md` for why).
- **ArchiLudo:** this project – a native WIMP port for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. The house rules the 1992 original encoded by hand are now the configurable `LUDO_VARIANT_MEJN` preset among three curated variants (see `RULES.md`), and its own multi-decade thread continues in this README's Changelog above.

## Documentation

See `docs/` for the full project documentation: `ARCHITECTURE.md` (layering, directory structure, porting notes), `RULES.md` (the full player-facing rules manual), `GAME_LOGIC.md` (rules engine API), `BOARD_LAYOUT.md` (board geometry), `AI.md` (AI design), `QTM.md` (music/SFX manual and audio API), `TOOLS.md` (every standalone asset/build tool, its usage, and gotchas), `BUILDCHAIN.md` (toolchain, Makefile, distribution), `OSLIB.md` and `LIBARCHIE.md` (the libraries this project builds against); plus `riscos_wimp_reference.md` for the WIMP/SWI/ message API reference, and `CREDITS.md` for everything this project is built on or derived from.

## **Credits and licence**

See `CREDITS.md` for everything ArchiLudo is built on, ported from, or otherwise sourced from. Licensed under the GNU GPLv3.